Mathematical Design of a Legal-Action Alakazam Controller

Exact card-count control under partial information

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Scope. This compact Kaggle template explains the controller through its mathematics and architectural invariants. The measurements below test semantic distinguishability, offline Prize prediction, and AWR weight stability; they do not prove a causal gameplay gain. The document has not been submitted.

Abstract

Powerful Hand Alakazam turns Pokémon TCG play into a coupled counting and control problem: hand size determines attack output, Prize values determine the shortest winning route, and every optional draw changes deck-out risk. The agent must solve that problem inside a variable, multi-stage legal action space without reading hidden opponent information. We formulate the final design as a constrained direct policy. A pinned simulator supplies the legal set; an autoregressive decoder assigns probability to every ordered legal selection; a public-rule semantic key prevents distinct choices from sharing one representation; and a bounded residual augments, rather than replaces, the learned option embedding. Training adds a masked three-action Prize-plan advantage, while serving remains direct-policy inference. The resulting design connects deck mathematics to model mathematics: exact knockout thresholds, safe hand slack, forced-draw feasibility, and Prize routes become causal quantities that the controller can represent and learn.

1. Constrained legal-action control

Let $I_t = \Pi_{\mathrm{act}}(o_{\le t})$ be everything visible to the acting seat: its own hand and private state, the public board and history, and the current simulator menu. Opponent hand identities, either deck order, unrevealed Prize identities, terminal results not yet observed, and future transitions are excluded. The pinned simulator, not a learned rule model, defines

$$\mathcal{A}_t = \mathcal{A}(I_t).$$

A prompt with n options and legal selection size $\ell \le k \le u$ contains

$$|\mathcal{A}_t| = \sum_{k=\ell}^{u} P(n,k) = \sum_{k=\ell}^{u} \frac{n!}{(n-k)!}$$

ordered complete actions. Enumerating that set is unnecessary. From prefix p, the decoder considers every unused extension $p \,\|\, i$ and, once $|p| \ge \ell$, an explicit STOP. At $|p| = u$, the prefix is complete. Thus, for $a = (i_1, \ldots, i_k)$,

$$\pi_\theta(a \mid I_t) = \prod_{j=1}^{k} \pi_\theta(i_j \mid I_t, p_{j-1}) \; \pi_\theta(\textsc{stop} \mid I_t, p_k)^{\mathbf{1}[k<u]}.$$

Every stage is masked to its simulator-supplied candidate set $C_t(p)$:

$$\pi_\theta(i \mid I_t, p) = \frac{\mathbf{1}[i \in C_t(p)] e^{\ell_i}}{\sum_{j \in C_t(p)} e^{\ell_j}}.$$

This factorization preserves complete legal support and recomputes legality after each selection, including order-sensitive effects.

2. Semantic quotient and bounded policy design

Legality alone does not guarantee distinguishability. Let $\kappa_0(I, o)$ be the legacy option key and $\sigma(I, o)$ the simulator semantics of choosing o. Its collision class is

$$C_k = \{(I, o) : \kappa_0(I, o) = k\}.$$

A collision is actionable exactly when there exist $x, y \in C_k$ with $\sigma(x) \ne \sigma(y)$. Duplicate payloads that differ only by a harmless permutation are one semantic equivalence class, not an error.

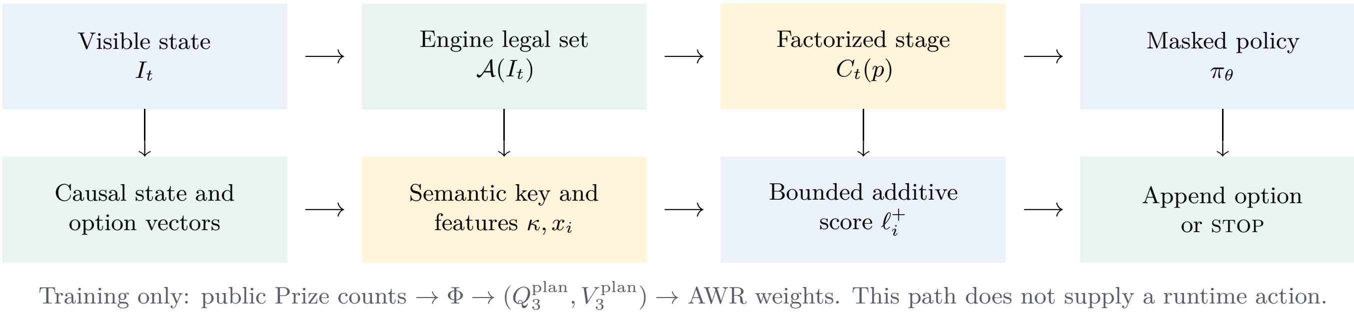

Figure 1: Mathematical and architectural boundary of the direct controller.

The repaired key κ uses acting-player-visible fields that can alter the meaning of a choice: exact number and count, selection type and context, effect identity, minimum and maximum count, remaining damage-counter and Energy budgets, and stable source, target, area, and slot bindings. It includes no candidate ordinal, blind global serial, card prose, or future outcome. Reordering non-semantic menu options therefore only reorders their rows.

The base Transformer [4] produces option state h_i and typed state/option predictions. A compact fusion summary can be written as

$$c_t = \mathrm{LN}\left(\frac{1}{m}\sum_{r=1}^{m} \alpha_r \tanh(W_r s_r)\right), \qquad \ell_i^0 = \mathrm{MLP}([h_i, c_t, \tanh(q_i)]).$$

Here s_r denotes typed state heads and q_i option-conditioned heads. The rule-semantic path is parallel to the preserved card embedding:

$$h_i^+ = h_i + g f_\psi(x_i), \qquad \ell_i^+ = \ell_i^0 + g\Delta \tanh(r_\psi(h_i^+)), \quad \Delta = 0.25.$$

The final layers begin at zero. When $g = 0$, the implementation returns the baseline tensor by identity before inspecting semantic features. The adapter therefore cannot invent legality and is not a second rules engine; it only separates meanings already exposed by the simulator. Unsupported checklist and metadata routes stay at exact zero.

3. Public Prize potential and H3-shaped AWR

Terminal win/loss is mathematically correct but sparse. The revision-27 design [2] instead fits a training-only monotone potential from public remaining-Prize counts:

$$\Phi(p_o, p_x) = 2P_{\text{iso}}(\text{win} \mid p_o, p_x) - 1.$$

If our remaining count falls, Φ may not fall; if the opponent's falls, Φ may not rise. For a fixed γ, the shaped segment reward and h-segment return are

$$r_t^P = \gamma\Phi(s_{t+1}) - \Phi(s_t), \qquad G_t^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^P = \gamma^h \Phi(s_{t+h}) - \Phi(s_t).$$

The sidecar target is $G_t^{(h)}/(1+\gamma^h)$. Missing, ambiguous, or terminal-censored segments are masked; they are never relabeled as zero.

The actor uses exactly one cumulative horizon:

$$A_{\text{H3}}(s,a) = z - V_{\text{existing}}(s) + 0.025\, m_3 c_3\, \mathcal{S}_{\text{train}} \left[Q_3^{\text{plan}}(s,a) - V_3^{\text{plan}}(s)\right].$$

$m_3 = 1$ only for three complete, causal, unambiguous segments from the same seat. $c_3 \in [0,1]$ is a frozen public confidence and action-support scalar, and $\mathcal{S}_{\text{train}}$ is frozen train-split scaling. H1, H6, and H12 have zero actor coefficient. The same complete action advantage is broadcast to its selected factorized stages, with no gradient into the sidecar.

Following AWR [1], optional batch whitening gives

$$\widehat{A}_i = \frac{A_i - \mu_B}{\max(\sigma_B, 10^{-6})}, \quad w_i = \min\{w_{\max}, e^{\widehat{A}_i/\beta}\}, \quad \mathcal{L}_\pi = -\frac{1}{|B|}\sum_i \mathrm{stopgrad}(w_i) \log \pi_\theta(a_i \mid I_i).$$

β and $w_{\max}$ remain configuration variables; no extra raw H3 advantage clip is introduced. The H3 model changes training weights only. It is absent from policy state, serving, and Kaggle inference.

4. Alakazam as constrained card-count optimization

The deck guide [3] supplies an unusually clean control variable. Powerful Hand places two damage counters per card in hand, so, when no visible effect blocks the attack,

$$D(h) = 20h, \qquad h_{\mathrm{KO}}(H) = \left\lceil \frac{H}{20} \right\rceil.$$

Ordinary Weakness and Resistance do not alter placed counters. A visible effect that prevents attack effects makes the route infeasible until removed or avoided. Against a full-health 330-HP Mega Lopunny ex, $h_{\mathrm{KO}} = \lceil 330/20 \rceil = 17$; the visible card class also makes the knockout worth three Prizes.

Let h be the current hand. The maximum hand slack that can be spent without losing the present knockout is

$$b_{\mathrm{spend}} = \max(0, h - h_{\mathrm{KO}}).$$

That scalar is necessary, not sufficient: spending a card can still destroy the next attacker or recovery line. For a planned route r, let D be the visible current deck count, u_r unavoidable draws, v_r publicly guaranteed recycled cards, ρ_r indicate a Benched Alakazam with Psychic-providing Energy, b_r occupied Bench slots, and B the Bench maximum. The default feasible set is

$$\mathcal{F}(I_t) = \{r : \ h_r \geq h_{\mathrm{KO}}(H_r),\ u_r \leq D + v_r,\ \rho_r = 1,\ b_r \leq B - 1\}.$$

The Bench reserve may be released for an already exact winning line. If play must continue to another turn, draw continuity strengthens the constraint to $u_r + 1 \leq D + v_r$ for the next mandatory draw. Equality in the first constraint is safe only when the route ends before that draw.

Let $q_j \in \{1, 2, 3\}$ be the visible Prize value of target j and P_{need} the Prizes still required. The primary route problem is

$$T^* = \min_{r \in \mathcal{F}(I_t)} T(r) \quad \text{subject to} \quad \sum_{j=1}^{T(r)} q_j \geq P_{\mathrm{need}}.$$

Among equal-turn routes, the guide prefers fewer forced draws, less two-Prize Bench exposure, and preservation of recovery. Deterministic search precedes broad draw because removing a known target changes the remaining distribution without spending unnecessary deck depth. Hidden Prize identities are never guessed: a route that requires an unknown key card is rejected until public evidence makes it feasible.

5. Mathematical checks

The main representation check used 18,412,973 option records from 2,351,208 decisions. The legacy key merged 255,398 records into 127,641 actionable collision groups, affecting 1.387% of all option rows. The repaired quotient left 99 duplicate groups; simulator witnesses classified every one as a harmless permutation, leaving zero unresolved actionable groups.

The H3 sidecar was evaluated offline on 380,499 complete validation actions. Exact three-segment targets existed for 358,344 actions (94.18%). Its action-value MSE was 0.0040745 versus 0.0046184 for the empirical-mean baseline, an 11.78% reduction. This is prediction evidence, not a win-rate claim. On the same 2,146,670-decision replay membership, adding H3 left the AWR weight distribution essentially unchanged: effective-sample-size fraction moved from 0.251337 to 0.251758 and clip fraction from 0.076078 to 0.075819.

6. Limits

The collision census proves representational separation on the sealed corpus, not universal rule coverage. The H3 results prove offline prediction and weight-distribution non-regression, not causal gameplay strength. Only the public-rule semantic projection is census-supported; checklist-provenance, unvalidated metadata, and repaired auxiliary branches remain zero or trace-only. A single leaderboard score is intentionally omitted because it cannot identify which mathematical component caused an outcome.

Table 1: Component-level evidence for the mathematical design.

Object	Measured check	Supported conclusion
Semantic key	255,398 legacy rows; 0 actionable repaired groups	Publicly different simulator meanings are separated
H3 prediction	94.18% coverage; Q_3 MSE 11.78% below mean baseline	Offline Prize-plan signal is learnable
AWR weights	ESS 0.251337 → 0.251758; clip 0.076078 → 0.075819	Bounded H3 term preserves replay-weight stability
Runtime boundary	H3 sidecar excluded; illegal logits masked	Direct policy retains engine-owned legality

Conclusion

The design treats play as legal constrained optimization: factorize the exact simulator action set, quotient options by public rule semantics, bound every additive score path, shape learning with causal Prize progress, and enforce Alakazam's hand, draw, replacement, Bench, and Prize inequalities. That is the core of the system; infrastructure is supporting evidence, not the story.

References

[1] X. B. Peng, A. Kumar, G. Zhang, and S. Levine. Advantage-weighted regression: Simple and scalable off-policy reinforcement learning. *arXiv preprint arXiv:1910.00177*, 2019.

[2] Project Team. Alakazam elmo rule-derivative contract, revision 27, 2026. Internal typed design authority; source identity recorded in the manifest.

[3] Project Team. Alakazam new-list direct-policy pilot guide, 2026. Internal exact-list mathematical strategy source; source identity recorded in the manifest.

[4] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.